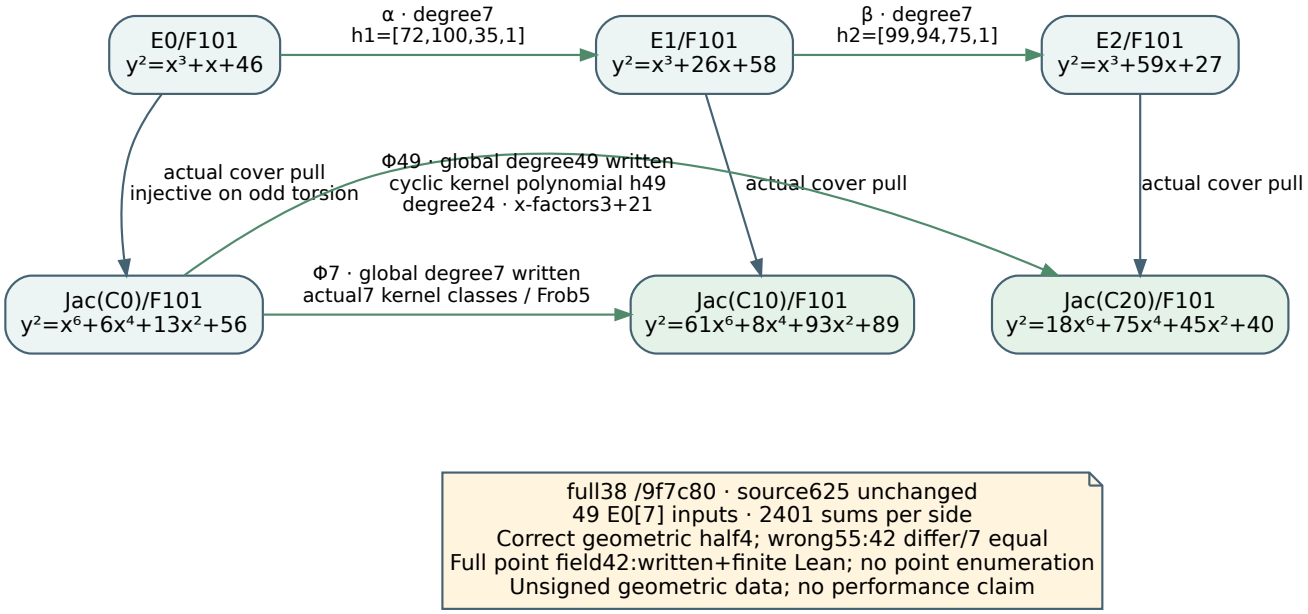
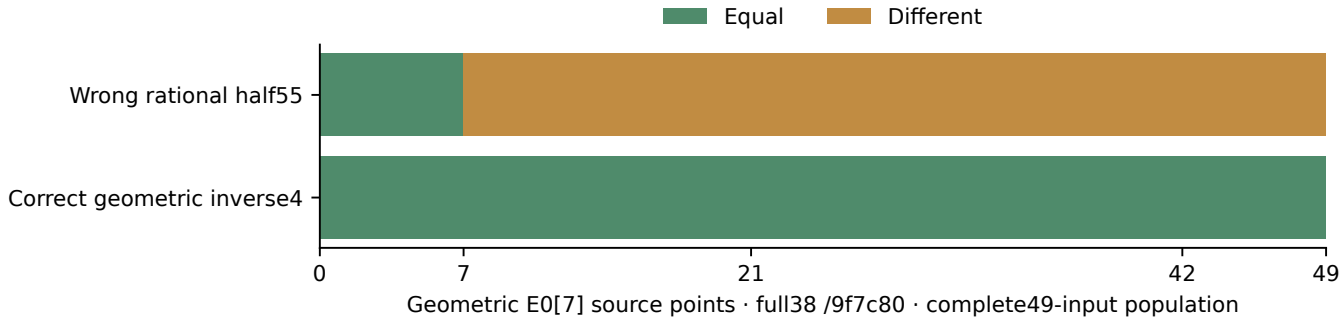
Actual seven-kernel and cyclic-49 certificates

2026-11-12 · full38 /9f7c80 · native + written + faithful Lean · unsigned

Complete38 native tests pass,zero failed/ignored/filtered;old reference+legacy guards,fmt,strictClippy and all625 input hashes pass. Deterministic24475-byte native JSON is byte-identical to diagnostic19142f. Five new bound local Lean passes yield188 written declarations/37 modules,zero trusted native. Algebra02/ d7d12d passes23 tests/eight outputs/all12 hashes;old seven outputs remain byte-identical. Scientific standing remains unsigned. [pcc:unverified]



Actual elliptic maps and cover pulls;global Jacobian degrees use written 128eb1/afd164. Native actual7 kernel and49-polynomial certificates have separate domains. Full point field42 is written/finite-Lean reasoning; no F101^42 point enumeration. [pcc:unverified]



Complete49-input E0[7] control against actual target pulls,no sampling uncertainty. Correct inverse4 agrees on49;wrong rational half55 agrees on the7 kernel points and differs on42. These are counts,not costs. [pcc:unverified]

All49 E0[7] points produce distinct valid C0 Mumford pulls with actual norm (2P,O),7-annihilation,all2401 source and target sum comparisons and both Frobenius commutators. Actual7 kernel classes satisfy all49 sums,nonzero order7 and Frobenius=[5]. This is the first-factor49-point slice,not the entire2401-point Jacobian[7]. [pcc:unverified]

Exact polynomial and field boundary

For actual α x-map N/D with D=h1²,and second kernel h2=[99,94,75,1],the complete degree24 kernel polynomial is h49=h1·D³·h2(N/D):

[42,52,3,51,36,99,66,64,34,90,39,51,50,4,39,23,77,42,5,55,33,49,35,17,1].

Native checks:monic,squarefree,coprime to x³+x+46,exact division of the degree1200 psi49 with a retained full quotient,and gcd(h49,psi7)=h1. The written fibre proof accounts for D=0,D≠0 and origin. The49-point composite kernel intersects E0[7] in7 points,so it is cyclic. Injective odd-torsion pullback and global degree49 identify the entire Phi49 kernel. [pcc:unverified]

Native repeated Frobenius degree d	gcd(h49,X^(101^d)-X)
1	1
3	h1
7	1
21	h49

These establish irreducible x-factor degrees3+21. Cyclic Frobenius scalar33 comes from lambda mod7=5 and T²+7T+101 modulo49. Old CyclicFortyNine proofs give unique33/order42;new x-period proof gives33²¹=-1 and no earlier ±1. Thus primitive49 points have x-degree21/full-degree42. Order7 kernel points have x-degree3/full-degree6. Actual cyclicity/elliptic characteristic polynomial/coordinate-field identifications remain written. [pcc:unverified]

Faithful formal map

EvenKernels.cubic_homogenization/c9aa3b is universal in a commutative ring. geometric_half_coordinates/e1bd7f and geometric_wrong_half_counts/f5bb27 cover actual finite scalar/count arithmetic;first_kernel_x_period/063ed6 and first_primitive_x_period/7310b6 cover exact periods. All5 bindings locally pass;188written37modules,0trusted. New IDC1ha31d00e9a1842a62 uses the unchanged checker at32+64 fixed2^61-1 points;it is not an F101 theorem. Every one of eight RHS+1 controls rejects. [pcc:unverified]

EVEN_KERNEL_CERTIFICATE_PROOF.md/063773 gives all global-kernel,cyclicity and field-degree bridges. Full raw129135 bytes→standardXZ11128,source/ start/finish/exit/afterhashes/certcmp/raw+archive SHA256/exactroundtrip are retained. The24475-byte native JSON includes field modulus,sextics/chart, full map/polynomial/point/kernel/sum/Frobenius data. [pcc:unverified]

Interrupted full02/eb1ebb and algebra01/2cb5c1 are not promoted despite test-pass stdout;original outer exits were lost. Initial protocol standing finding and wrong CLI path are also retained in the separate failure report. Complete full03/algebra02/gate03 replace those invocations without source or criterion changes. No native mathematical prediction failed. [pcc:unverified]

Sources:128eb1/afd164,unchanged native engines,Bruin–Doerksen0902.3480 Lemmas2.4/2.6,Galbraith corrected9.8–9.9. KB unavailable/exact fallback, no novelty inference. LocalLinuxx86_64 workspace unit checks,no benchmark isolation qualification. Rust1.99.1/Lean4.35.0. Own new CI follows this commit; 214 compiled expected,not yet own-CI observed. Prior d87 CI checks19:06UTC. Next:primitive-four/T20 + independent CT49 Richelot validation,additive consumer replay,admitted full costs. No paid run or benchmark. REPORT.md links source/custody/editable figures/data/renderer and this PDF. [pcc:unverified]